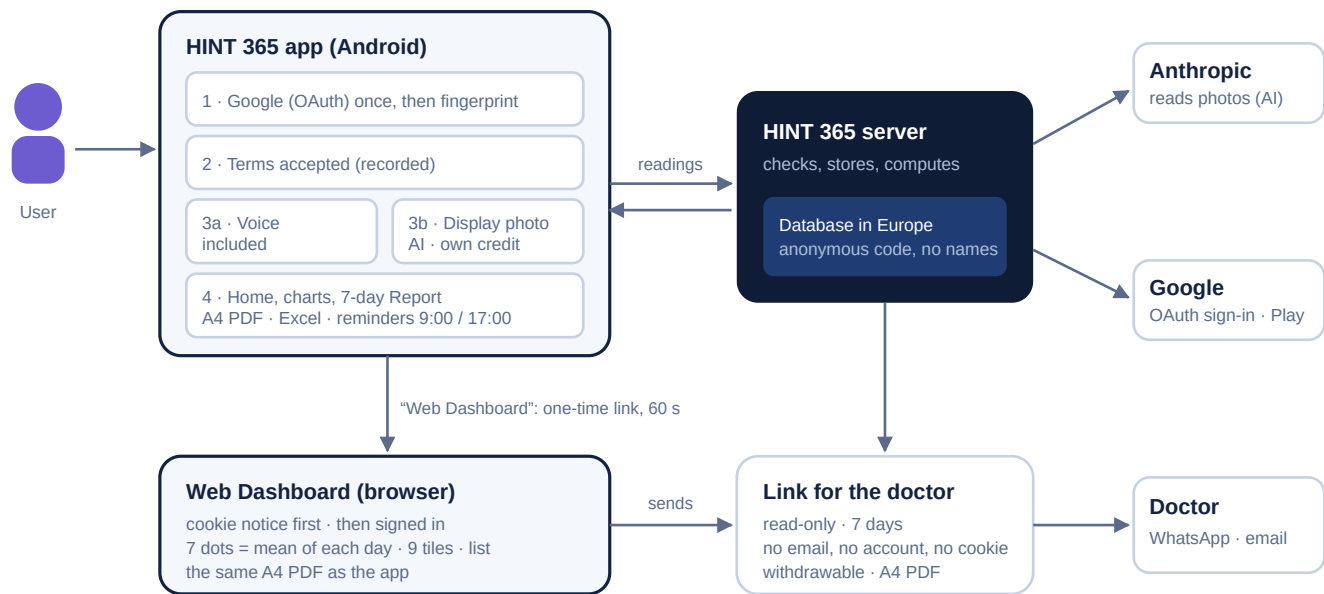


Functional overview

What the app does, who does what and in which order. A blood-pressure diary for Android, with a web dashboard and reports for the doctor.

The flow from the phone to the doctor



The user's path

1. **Installs** the app from the public link `/download` (later from Google Play).
2. **Signs in with Google** once, through OAuth 2.0 / OpenID Connect (a signed ID token, never the password), and gives privacy consent.
3. **Accepts the terms of use**, binding, at the first installation and after every update: without acceptance the app does not open. Every acceptance is recorded (anonymous code, phone, versions, text fingerprint, date).
4. **Every day** unlocks with the phone's fingerprint or PIN, and stays signed in until "Sign out".
5. **Records** a reading: by voice (**SYS**, **DIA**, **PUL**, checked and confirmed on screen) or by photographing the display (AI features).
6. **Looks back**: last reading, the week, the Report with charts and tiles.
7. **Shares**: PDF or Excel from the phone, or Web Dashboard → "Send to the doctor".
8. **Changes phone**: signs in with Google and finds everything. **Signs out** or **deletes the account** at any time.

Costs and who is paid

What	Cost	Paid to
HINT 365: app, charts, reports, Web Dashboard, keeping the readings	5 \$ a year, cancel any time	The app's owner, through Google Play
AI features, optional: Scan of the monitor (available now); anonymous medical documents (coming)	Pay-per-use, ≈ 0.006 \$ a photo	Anthropic, from the user's own credit

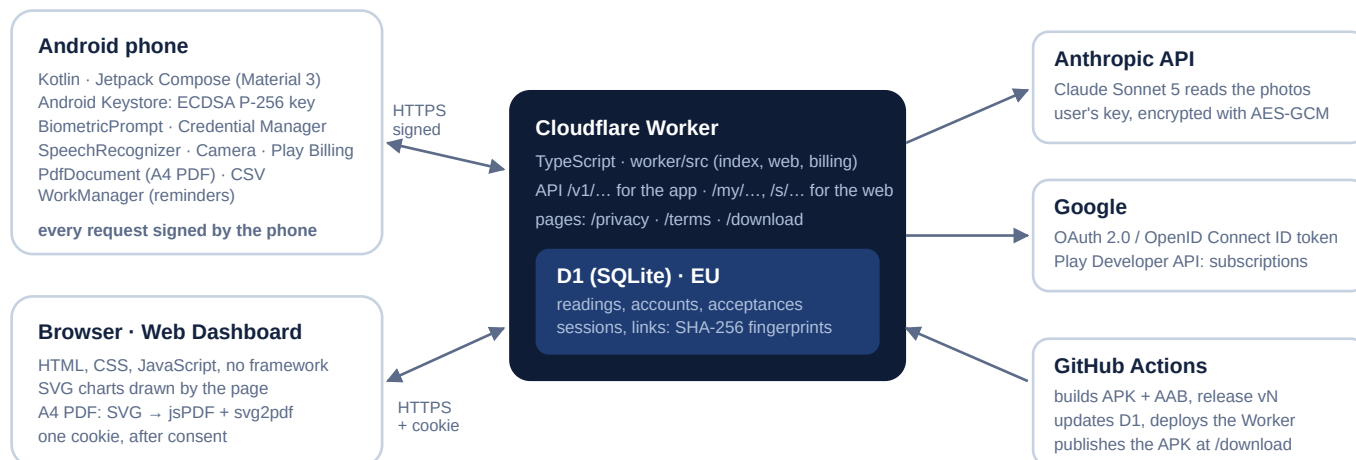
One app; the AI features are on or off (never "Premium"). The Anthropic credit is the user's own.

Fixed rules

- No diagnosis, no advice, no judgement of the values: no coloured zones, no "normal".
- Labels always **SYS**, **DIA**, **PUL**. Pressure and pulse never in the same chart.
- Charts: 7 days, one dot per day = the mean of that day's readings (explained before the charts), value next to it, axis with day numbers only.
- The same A4 PDF from the app and from the web; the web downloads it at once as a file.
- No red anywhere: **SYS** violet, **DIA** teal, **PUL** amber. No colour means "good" or "bad".
- Scan is on only when Anthropic has just said yes (valid key, credit): a minimal request on opening if the last check is over an hour old, after every error and when the screen is pulled down.
- Readings kept at most 365 days, deleted nightly; only the acceptances of the terms remain, as proof.

Architecture and technology

The components, how they talk, where the data live and how a new version is released.



Components

Part	Technology	Folder
Android app	Kotlin, Compose, minSdk 26, target 34	android/
Server	Cloudflare Workers, TypeScript	worker/src/
Database	Cloudflare D1 (SQLite), EU jurisdiction	worker/schema.sql
Web dashboard	HTML/JS, own SVG charts, jsPDF + svg2pdf	worker/public/my/
Photo reading	Anthropic Messages API	index.ts
Sign-in	Sign in with Google (OAuth 2.0 / OpenID Connect) + phone key	Google.kt
Subscription	Google Play Billing + Play Developer API	Billing.kt , billing.ts
Build, release	GitHub Actions, Gradle, Wrangler	.github/workflows/
Guides, documents	HTML → PDF (Playwright)	docs/

Database tables

persons · accounts (anonymous code, encrypted fingerprint of the Google account, no email; subscription sub_token , sub_until , sub_state) · measurements readings · scans photo readings · ledger credit and costs · person_keys encrypted Anthropic keys · acceptances accepted terms (append-only) · web_codes , web_sessions , web_shares web access and doctor links · settings , invites .

Extending

Modular: lab reports will be a new module in web.ts and app.js .

Security and data

- Phone:** a private key that never leaves the phone signs every request; the server refuses bad signatures or clocks off by more than 5 minutes.
- Web:** a 60 s one-time code in the # part of the address becomes the only cookie, hint_s (technical, 7 days, HttpOnly, Secure, SameSite=Strict), set only after the cookie notice is accepted; the choice stays in local storage. The app uses no cookies.
- Doctor links:** read-only, fixed period, expire after 7 days, withdrawable; no email, no account, no cookie.
- Remote switch-off:** the app version travels with every request; settings holds app_min_version , app_blocked , app_off . Owner only, from the web Admin area.
- Owner's area** (web, owner only): usage numbers per anonymous code (joined, last opened, readings, AI on), D1 size against the 500 MB free limit. Never readings, reports or names.
- Subscription** (ready, off until Google Play): Play Billing, checked with the Play Developer API.
- Database:** only SHA-256 fingerprints of codes, cookies and links; Anthropic keys encrypted with AES-GCM.
- Retention:** readings older than 365 days deleted every night; acceptances in an append-only register.
- Personal data:** no name, no email (of the Google account only an HMAC-SHA256 fingerprint; the email stays on the phone); photos and files processed and discarded; data in the EU.

Releasing a version

- Change on a branch → pull request → merge into main .
- GitHub Actions updates D1, deploys the Worker, builds the APK and the AAB, creates release v0.1.N and puts the APK at /download .
- The site changes at once; the app updates by installing the new APK.

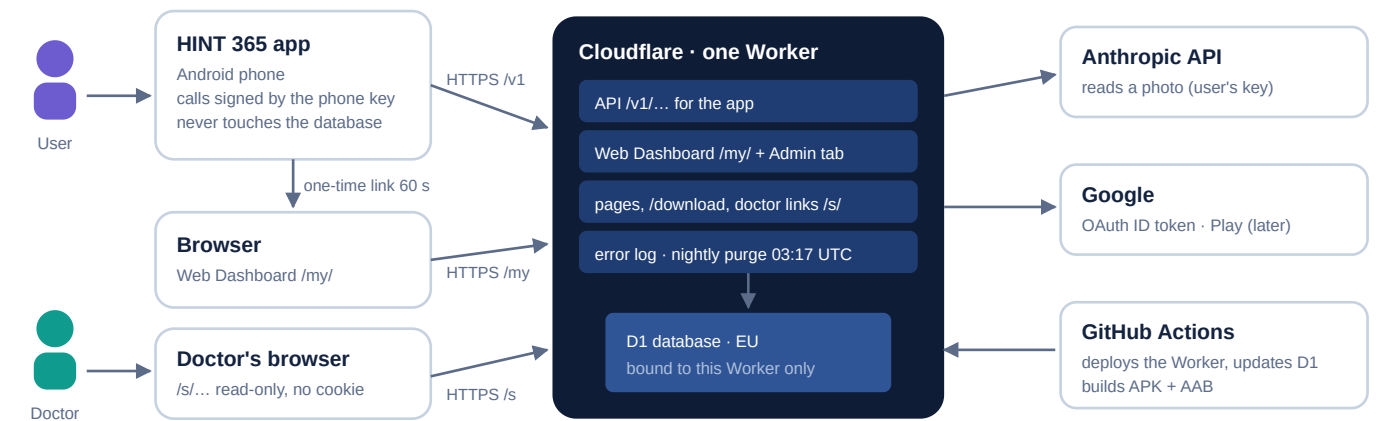
Hosting and runtime flow

Where every part runs, who talks to whom at run time, and how errors met by users are recorded.

Where it runs

Part	Runs on	Address	Operated by
Android app	The user's phone (APK; Google Play later)	installed from /download	the user
Web Dashboard (incl. owner's Admin tab)	Cloudflare Workers (static assets of the Worker) — not Vercel	personalhealthy-api.comevava76.workers.dev/my/	Cloudflare, owner's account
API for the app, public pages, doctor links	Cloudflare Workers, the same single Worker	.../v1/... · /privacy · /terms · /s/...	Cloudflare, owner's account
Database	Cloudflare D1 (SQLite), EU jurisdiction	no address: bound to the Worker only	Cloudflare, owner's account
Source, CI/CD, releases	GitHub: repository, Actions, Releases	comevava76-web/personalhealthy (private)	GitHub, owner's account

Runtime flow only the Worker touches the database



Error log continuous testing and improvement

- **Server:** every failed call to `/v1` or `/my/api` (status 400–599, except 402, 404, 426 and a browser without a session) and every exception.
- **App:** crashes (sent when the app opens again) and failures of local work such as building the PDF or the Excel file. At most 20 a session.
- **Web:** script errors of the Web Dashboard, at most 5 a page; never on a doctor's link.
- **Short by design:** one row per day, source, code, place and app version, with a counter. Digits are removed from messages, so no reading value is stored; only the anonymous account code. Kept 90 days.

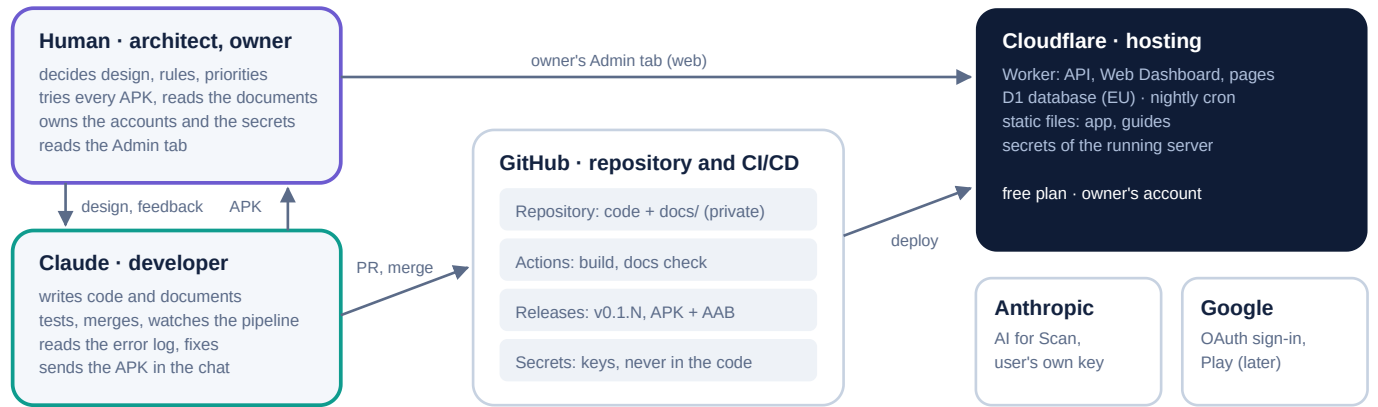
Who uses it

- **Owner:** Admin tab → “Errors of the last 30 days” and the 7-day count.
- **Developer:** every morning a scheduled routine reads the report of the log and fixes (page 4); in working sessions it is also read after every release.
- **Table:** `error_log` in D1, written by `worker/src/errors.ts`; app side `ErrorReport` in `Core.kt`; web side in `app.js`.
- Also on every account: the app version it last opened with and whether the AI features are on, for the owner's Admin tab.

Roles and delivery

Who does what — the architect, the developer, the repository and the hosting — and how a change goes from the chat to the phone.

Roles people and platforms



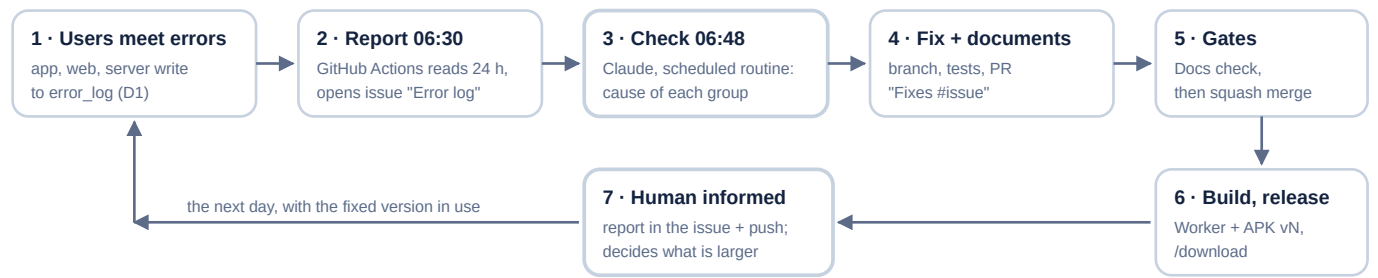
Responsibilities

Activity	Human · architect	Claude · developer	GitHub	Cloudflare
Design, rules, priorities	decides	proposes, asks when unclear	—	—
Code and documents (docs/)	reads	writes, same commit	stores	—
Tests before merge	—	runs	Docs check	—
Merge, build, release	delegated	merges, watches	builds, releases	—
Hosting, database, cron	owns the account	—	deploys	runs
Keys and secrets	owns	never sees them	stores	stores
Trying the app on a phone	tries every APK	sends it	—	serves /download
Errors met by users	sees them in Admin	analyses, fixes	—	stores the log

Delivery, from the chat to the phone

- 1. Decide** — Human describes the change in the chat; unclear points are asked, not guessed.
- 2. Build and test** — Claude on branch `cclaude/...`: code and documents in one commit; Worker bundle, pages in Chromium, documents rendered.
- 3. Merge** — pull request, *Docs check* green, squash merge into `main`.
- 4. Release** — GitHub Actions: D1 tables, Worker, APK/AAB `v0.1.N`, `/download`; Claude checks it and sends the APK.
- 5. Try** — Human tries the APK; feedback starts the next round. Rollback: revert the pull request.

Continuous improvement loop proactive: every day, without being asked



Small fixes are merged by the routine; larger ones, or the terms, wait for the human. Not automated yet: unit tests, real-phone tests.

Application flows and permissions

Every flow of the application, step by step, and who is allowed to do what.

Application flows

#	Flow	Who	Steps	Data
1	Install, sign in	user	/download → install → Sign in with Google (OAuth, ID token checked by the Worker) → phone key registered	persons (fingerprint only)
2	Accept the terms	user	shown at first install, after every update and when the text changes → accept or the app closes	acceptances
3	Record by voice	user	Record → say SYS, DIA, PUL → check → Save reading → /v1/bp/voice	measurements
4	Turn on AI	user	Anthropic account → credit → key → paste → Worker tests it, stores it encrypted	person_keys
5	Scan (AI)	user with AI on	Scan → photo → Worker → Anthropic → values → check → Save → /v1/bp/confirm; photo discarded	scans, measurements, ledger
6	See the report	user	Report tab: 7 days, one mean per day, morning/evening, extremes	read only
7	Share the report — app	user	Report → Send PDF or Excel → Android share sheet (WhatsApp, email, files...). The file goes straight from the phone to the app chosen; the server is not involved	none on the server
8	Share the report — web	user	Report → Web Dashboard → cookie consent → PDF download (same A4) or Send to the doctor : Email or WhatsApp only; the PDF and a read-only link (7 days) go together, with a ready message in the phone's language; withdrawable	web_sessions, web_shares
9	Doctor's view	anyone with the link	open /s/... → charts and readings of that period, PDF; nothing else	read only
10	Delete, sign out	user	Admin → delete one reading, all, or the account; Sign out also ends web sessions	deleted at once
11	Owner's Admin tab	owner	Web Dashboard → Admin: usage per anonymous code, app version, AI on/off, D1 size, errors, version switch-off	counts only
12	Automatic	system	nightly: readings > 365 days, expired links and sessions, errors > 90 days deleted; every error logged	purge, error_log

Who can do what

Action	User · app	User · web	Doctor · link	Owner	Developer
Record, scan, delete own readings	✓	—	—	own only	—
See own charts and report, export PDF/Excel	✓	✓ (PDF)	the shared period	own only	—
Create or withdraw a doctor link	—	✓	—	own only	—
See another person's readings or report	—	—	—	never	never
See usage numbers, app version, AI on/off per account	—	—	—	✓ Admin tab	via the owner's connector, for checks
Read the error log	—	—	—	✓ Admin tab	✓ to fix
Switch app versions off	—	—	—	✓ Admin tab / Actions	—
Change code, docs, deploy	—	—	—	decides	✓ through pull requests
See keys and secrets	own AI key, once	—	—	✓ owns them	never

The server enforces every line: app calls are signed by the phone key, web calls need the session cookie, owner calls need the owner account, doctor links are read-only and expire.